

Ch-3 — Heat

Science — Physics • Class 7 • Max Marks: 17 • Time: 45 min

Name: _____ Roll No: _____ Date: _____

Topics covered: Hot & cold and temperature, measuring temperature, clinical and laboratory thermometers, transfer of heat (conduction, convection, radiation), conductors & insulators, sea & land breeze, dark/light surfaces.

Section A — Multiple Choice (Higher Order) (1 × 14 = 14)

1. Two identical cups of hot water are taken; one is wrapped in a woollen cloth. After 10 minutes, the water that is hotter is in the:

- A. neither — both are equal
- B. cup wrapped in wool
- C. bare cup
- D. cup that was stirred

2. On a cold morning an iron bench and a wooden bench are at the same temperature, yet the iron feels colder. This is because iron:

- A. conducts heat away from the hand faster
- B. is heavier
- C. radiates cold
- D. is actually colder

3. Heated over the same flame for the same time, water gets hottest in a:

- A. shiny silver metal pot
- B. glass beaker
- C. shiny white ceramic pot
- D. dull black metal pot

4. The temperature of a body is best described as a measure of:

- A. its mass
- B. its volume
- C. the average degree of hotness of its particles
- D. the total heat energy stored in it

5. The shiny silvered walls of a thermos flask mainly reduce heat loss by:

- A. radiation
- B. conduction
- C. convection
- D. all three equally

6. Rods of copper, aluminium, iron and glass of equal size are heated at one end with a wax ball at the other. The ball that falls LAST is on the:

- A. iron rod
- B. glass rod
- C. copper rod
- D. aluminium rod

7. During the day a sea breeze sets in because the air over the land:

- A. becomes cooler
- B. gets warmer and rises
- C. sinks down
- D. becomes denser

8. Loose white cotton clothes are comfortable in summer because white reflects heat and loose cotton also:

- A. soaks up sweat and lets air flow (convection)
- B. radiates heat inward
- C. blocks all air
- D. conducts heat to the body

9. Which arrangement keeps a block of ice from melting the longest?

- A. ice in a metal box
- B. ice wrapped in a woollen blanket
- C. ice on an open steel plate
- D. ice on a black tray in sunlight

10. The only mode of heat transfer that can take place through a vacuum is:

- A. conduction and convection
- B. convection
- C. radiation
- D. conduction

11. A laboratory thermometer generally has a range of about:

- A. 0°C to 50°C
- B. 100°C to 250°C
- C. 35°C to 42°C
- D. -10°C to 110°C

12. The reading of a clinical thermometer does not fall on its own when taken out of the mouth because:

- A. the glass is thick
- B. mercury is very heavy
- C. the kink stops the mercury flowing back
- D. it is a digital device

13. A patient's temperature is recorded as 40°C. Compared with the normal body temperature, this shows a fever of:

- A. 3°C above normal
- B. no fever
- C. 3°C below normal
- D. 40°C above normal

14. In an experiment, equal wax balls on copper, iron and glass rods fall at 20 s, 45 s and 90 s. The best conductor of heat is:

- A. all are equal
- B. iron
- C. glass
- D. copper

Section B — Assertion–Reason (1 × 3 = 3)

Choose: (A) both A and R true and R explains A; (B) both true but R does not explain A; (C) A true, R false; (D) A false, R true.

1. Assertion (A): Metals are used to make cooking utensils.

Reason (R): Metals are good conductors of heat.

2. Assertion (A): On a sunny day, a car with dark-coloured seats becomes hotter inside than one with light-coloured seats.

Reason (R): Dark surfaces are good absorbers of heat radiation.

3. Assertion (A): Birds often fluff up their feathers in cold weather.

Reason (R): The air trapped between the feathers is a good conductor of heat.

Answer Key

Multiple Choice (Higher Order)

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. B | 2. A | 3. D | 4. C | 5. A |
| 6. B | 7. B | 8. A | 9. B | 10. C |
| 11. D | 12. C | 13. A | 14. D | |

Assertion–Reason

- 1. A
- 2. A
- 3. C